

(3 Hours)

[Total Marks : 100

N.B. : (1) Question No. 1 is **compulsory**.(2) Assume suitable **data** wherever **necessary**.(3) Solve any **four** out of the **remaining**.

1. (a) Using Quine Mccluskey method implement the following using gates 10
 $F(W, X, Y, Z) = \pi_m (1, 3, 4, 6, 9, 10, 11) * d(5, 8)$
 - (b) State De' Morgans theorem and prove that NAND gate is an Universal gate. 10
 2. (a) Simplify using boolean theorems and draw logic diagram for the same 10
 (i) $\bar{X} Y Z + X \bar{Y} Z + X Y \bar{Z} + X Y Z$
 (ii) $\bar{W} Y + \overline{\bar{W} X Z} + \bar{W} X \bar{Y} \bar{Z} + W \bar{X} Y$
 - (b) Realise the following using 16 : 1 MUX and 4 : 16 decoder. 10
 $f(A, B, C, D) = \Sigma_m (1, 2, 4, 7, 11, 13)$
 3. (a) Explain why Hamming code is called an error correcting code. Prove the same 10
 if a 7 – bit hamming code is received as 1 0 0 0 0 1 0.
 - (b) Design a 4 bit adder (BCD adder) using IC7483 10
 4. (a) Design a decade counter using JK flipflop. 10
 - (b) Design a mod 13 down counter using JK flipflop. 10
 5. (a) Minimize the following equation – 10
 $F(W, X, Y, Z) = \Sigma_m (0, 4, 5, 6, 12, 13) + d(7, 15)$
 (i) Design k – map for the same
 (ii) Minimize and realize the equation via gates.
 - (b) Explain what is meant by shift register. Explain 4 bit bi directional shift register. 10
 6. (a) Compare the different logic families with respect to following parameters – Fan in, fan out, noise margin, speed, power dissipation. 10
 - (b) Explain 2 input TTL NAND gate in detail. 10
 7. Write short notes on any **two** of the following :— 20
 - (a) Self complementry code
 - (b) PAL, PLA
 - (c) ALU
 - (d) 74180 – parity generator and checker.
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